

ReVerie: Contemplating Agency and Truth in AI-assisted Immersive Dream Reliving

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Figure 1: One audience member whispering her dream object into the *ReVerie* AI system.

ABSTRACT

ReVerie AI installation explores the scientific concept of dream reliving. Visitors engage in a ritualistic act of whispering their dreams. Each dream transcription and corresponding visualization are temporarily stored in the memory of the AI system. After that, the *ReVerie* AI system makes an inference from this memory to generate a shared, evolving A/V experience, where participants fly through interconnected dream objects in a virtual space. At its core, *ReVerie* challenges the capacity of technology to authentically represent the intangible and private terrain of dreams. By inviting participants to contribute whispered dreams that are algorithmically transformed into 3D dream objects, *ReVerie* pushes them to confront whether such digitally synthesized representations can capture the true essence of our subconscious—or if they craft new forms of detachment or control over personal narratives instead.

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1 DESCRIPTION OF ARTWORK

1.1 Introduction

ReVerie installation is a deeply reflective exploration of dreamwork through 3D generative AI, merging the personal with the communal in a shared digital landscape. In dream science, “Dreamwork” encompasses practices such as dream analysis and exploration aimed at uncovering insights into the subconscious mind. Central to this is the re-experiencing or reliving of dreams—immersing in memories and emotions to uncover subconscious meanings. The *ReVerie* AI system explores the scientific concept of “dream reliving”. It translates whispered dream recollections into real-time visualizations of dream objects using a 3D generative diffusion model, allowing participants to explore a shared “dreamscape”.

At its core, our artwork challenges the capacity of technology to represent the intangible and private terrain of dreams authentically. By inviting participants to contribute whispered dreams that are algorithmically transformed into 3D visuals, the artwork transforms dreams into a public, collective experience, pushing viewers to con-

front whether such digitally synthesized representations can capture the true essence of our subconscious—or if they instead craft new forms of detachment or control over personal narratives.

Historically, dream interpretation has often been wielded as a tool of power, such as in imperial China, where dreams were used to legitimize political ideologies, or in Mesopotamia and Egypt, where they were seen as messages from divine realms, shaping collective beliefs and societal behavior. This parallels our time, where state actors deploy mass social engineering to shape the collective subconscious with synthesized content and targeted misinformation through generative AI and social media. This modern approach continues the age-old use of subconscious influence, now heightened by digital means that permeate individual and societal realities.

Through this juxtaposition, *ReVerie* underscores how modern AI, with its generative and interpretive capacities, can either challenge or perpetuate subconscious control. By allowing participants to co-create in an open dreamscape, *ReVerie* offers an alternate route—a reclaiming of agency, where individuals contribute to a living, evolving collective dream that resists singular interpretations.

1.2 Experience

During the exhibition, visitors engage in a ritualistic act of whispering their dreams to the system. Each dream transcription and its associated visualization are temporarily collected in the memory of the AI system. Every five minutes, *ReVerie* retrieves from this memory to create a collective, dynamic visual journey where participants navigate through interlinked dream objects within a virtual space. The crafted soundscape delivers a range of emotional tones and social interactions to enhance the immersive auditory experience. This alignment of emotions tries to embrace individual interpretations, shaped by personal and cultural backgrounds, thereby enriching each unique dream scenario with emotion-specific music compositions.



Figure 2: The audience participates in a collective dreaming experience with their 3D dream objects generated in real-time.

1.3 Implementation

As participants whisper their dream descriptions into the system, a speech recognition module transcribes these into text, which then prompts a generative diffusion model to produce 3D point cloud data. This data is rendered in Unity, creating immersive dream visuals that blend individual dream memories with digital replication. The composer used generative music techniques to efficiently explore musical and timbral possibilities to reflect dream emotions at an early composition stage. Using Fast Fourier Transform analysis, the machine listening system segments audio and then computes each segment's emotion characteristics using music emotion recognition techniques. The composer then uses valence and arousal to validate the emotion category and adjust the music composition iteratively depending on the computational results of emotional data.

1.4 Reflection

The interplay of digital and personal reality within the installation intensified through ritualistic whispering and dream visualizations, urges participants to consider the authenticity of participatory experiences with generative AI. By actively engaging in this process, viewers question the truthfulness of these digital representations, examining whether they liberate or constrain the subconscious. Through this communal interaction, *ReVerie* also touches on the broader theme of identity and memory in the digital age: Can technology offer genuine insight into our inner worlds, or does it inevitably transform our dreams into artifacts, shaping a new kind of reality that is both ours and not ours?

In navigating this latent space between individual memory and digital fabrication, *ReVerie* questions our conceptions of the self and invites deeper contemplation of agency and truth in the digital realms of subconscious experience. Such immersive experience reflects the evolving relationship between human and machine, dream and reality, in a time when digital representations are ever more entwined with our identities.

2 ARTIST INFORMATION

Pinyao Liu is an interdisciplinary artist and researcher based in Linz, Austria, and Vancouver, Canada. His art practice and research explore generative AI and immersive technology for altered states of consciousness and transformative experiences. His interactive artwork has been exhibited at the 13th Shanghai Biennale at Shanghai Contemporary Art Museum, Vancouver International Film Festival VR Gallery, SIGGRAPH Immersive Pavilion and Ars Electronica Festival. His artistic research has been published in the ACM Conference on Human Factors in Computing Systems (CHI) and the Leonardo Journal by MIT Press.

Keon Ju Maverick Lee is a multidisciplinary artist-researcher, percussionist, and composer specializing in designing co-creative musical AI agents and interactive systems for drum and percussion performances. Currently pursuing a Ph.D. in the Metacreation Lab for Creative AI at Simon Fraser University, Keon integrates academic research with artistic practice to explore the intersection of music and artificial intelligence. His custom AI system, enabling dynamic collaboration between human performers and AI, was featured in Echoes of Synthetic Forest, a Top 10 finalist in the 2024 AI Music Song Contest, and showcased at the MusicAcoustica Festival and Montréal's Society for Arts and Technology during the ArtIA Symposium. He has collaborated with researchers at global institutions like IRCAM in Paris, CIRMMT in Montréal, and C4DM in London and published in esteemed venues such as the NIPS, AI Music Creativity Conference and Transactions of the International Society for Music Information Retrieval.

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